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# Learning to Select Local-Search Operators for EVRP-TW

A seed-separated Double DQN study with independent feasibility validation

## RESEARCH QUESTION

**Can a compact learned policy choose local-search operators more effectively than a fixed schedule or UCB1 on held-out EVRP-TW instances?**

**A feasible solution must**

- **Serve every customer** within its time window
- **Respect capacity** and battery limits
- **Use charging stations** when energy is insufficient
- **Return to the depot** with complete customer coverage

**Scope: hybrid RL controls local search after route construction. It does not build routes end to end.**

## METHOD OVERVIEW

EVRP-TW instance → Candidate portfolio Nearest + composite → Feasible 2-opt warm start → Operator selector Fixed, UCB1, Double DQN → Independent validator → Best feasible solution

**Every accepted move must reduce distance and pass the full EVRP-TW validator.**

**STATE**
**12 values**
scale, fleet use, progress, stagnation, prior action

**AGENT**
**12 inputs, 32 hidden units, 3 Q-values**
Double DQN with replay, target network, Huber loss

**ACTIONS**
**2-opt, relocate, swap**
12 decisions or stop after 4 non-improving moves

The learned policy selects one existing deterministic operator at each step; the solver retains only independently validated improvements.

## EXPERIMENT DESIGN

### 3 profiles × 3 scales

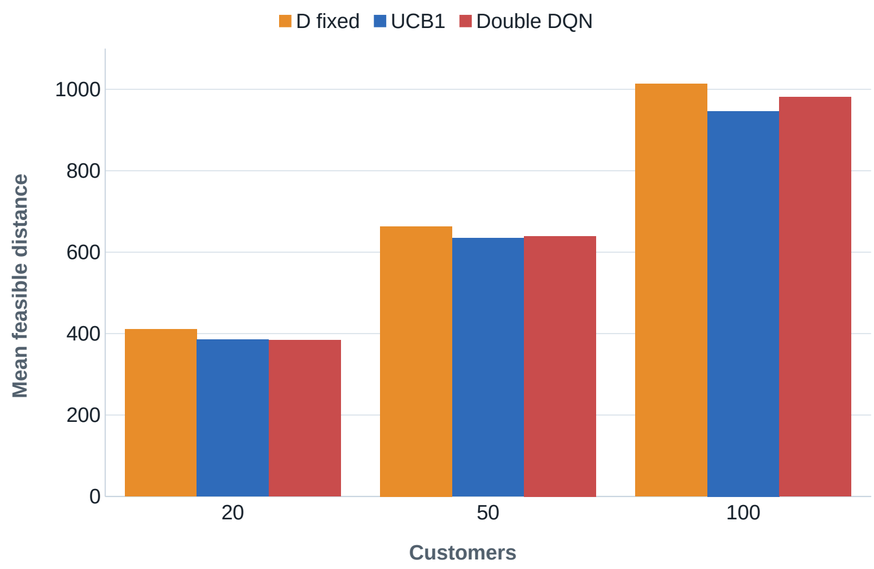
| Profiles | Customers |
|---|---|
| Baseline | 20 |
| Tight time windows | 50 |
| Small battery | 100 |

### Fair comparison

- Disjoint training and evaluation seeds
- Same instances, operators, validator, and search budget
- Feasibility and runtime include every attempted run
- Distance averages include feasible routes only

**Compared methods: D fixed, E fixed portfolio, UCB1, and Double DQN**

## HELD-OUT RESULTS




Baseline profile; lower distance is better. All four methods are feasible on all 18 baseline instances.

**Double DQN improves on the Week 5 method in every profile-scale cell, but does not consistently beat UCB1.**

- **1.62% to 8.20% shorter** than D fixed in all 9 cells
- **5 wins, 1 tie, 3 losses** against UCB1
- **Same feasibility** as each reference in every aggregate comparison
- **Baseline n=100** is 3.65% worse than UCB1

## GENERALIZATION, EVIDENCE, AND LIMITS

### Double DQN distance delta vs UCB1

| Profile | 20 | 50 | 100 |
|---|---|---|---|
| Baseline | -0.55% | +0.70% | +3.65% |
| Tight TW | -0.21% | -1.01% | -0.16% |
| Small battery | 0.00% | +0.43% | -0.95% |

Negative values mean Double DQN is shorter. The learned policy generalizes best under tight time windows, but loses at larger baseline scale.

**432** training episodes

**216** held-out method runs

### Same held-out instance, n=50




On this instance, the portfolio methods reduce distance from 679.5 to 626.3. UCB1 and Double DQN reach the same objective through different operator sequences.

**0** train-test seed overlap

**0** validation mismatches

### Conclusion and next step

**A small Double DQN can learn useful operator preferences while preserving feasibility, but the aggregate state does not fully represent route geometry.**

- One small-battery warm start is infeasible for every method.
- Current evidence uses synthetic instances from one generator.
- Next: add construction repair and a route-aware encoder, then test Schneider benchmarks.

Selected references
Schneider et al., Transportation Science, 2014
Lin et al., IEEE T-ITS, 2021
Kool et al., ICLR, 2019; Kwon et al., NeurIPS, 2020

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